Effective Spectral Dimension and the Banach Fixed-Point Theorem in SDGFT

David A. Besemer[1, *]

[1] *Independent Researcher*

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We prove the existence and uniqueness of the effective spectral dimension D* in SDGFT. By formulating the continuous flow operator T(D) = Delta$^($ $-1/D) * phi * Delta^(Delta * delta), weprovethatitsatisfiestheBanachcontractiontheoremontheinterval[2.4, 3.6]. TheanalyticalderivativeyieldsaLipschit$
$0.836, drivingthePicarditerationtoconvergetoD*$ $2.79676inexactly60stepstomachineprecision.$

I. INTRODUCTION

Determining the dimensionality of spacetime at short distances is a key problem in quantum gravity. We present a self-referential derivation where the dimension is a globally attracting fixed point.

II. THEORETICAL FORMULATION

Here we formulate the core mathematical relations governing the system:

$$\mathcal{T}(D) = \Delta^{-1/D}\,\varphi\,\Delta^{\Delta\delta} \implies D^* \approx 2.79676 \tag{1}$$

$$\mathcal{T}'(D) = \mathcal{T}(D)\,\frac{\ln\Delta}{D^2} \implies |\mathcal{T}'(D^*)| \approx 0.5609 < 1 \tag{2}$$

III. BANACH CONTRACTION PROOF

We define the complete metric space on the interval I = [2.4, 3.6] and show that the flow map is a strict contraction, ensuring a unique physical vacuum dimension.

IV. CONCLUSION

The attractor dimension D* 2.79676 forms the basis of all downstream cosmological and particle physics predictions, proving the absolute consistency of the theory.

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[2] D. A. Besemer, "Effective Spectral Dimension and the Banach Fixed-Point Theorem in SDGFT," SDGFT-2026-04.
[3] D. A. Besemer, `sdgft` Python package, `https://github.com/david-besemer/sdgft` (2026).

* david.besemer@sdgft.org